



**Faculty of Science and Arts
Department of Chemistry
Biochemistry Lab (CHEM 266)**

EXPERIMENT 9

GENERAL TESTS FOR CARBOHYDRATES

Pre-Lab Questions:

1- Classify the following sugars as reducing or non reducing:

Glycogen:

Lactose:

Xylose:

Glucose:

Starch:

2- What is the bond name between the monosaccharides in carbohydrates ?

Precautions

- Handle the chemicals with care.
- Heating should be done very carefully.
- Use droppers to take reagents from the bottles.

Objectives:

1. Study the various qualitative tests of carbohydrates.
2. Identify an unknown carbohydrate.
3. Explore the chemical and physical properties of carbohydrates.

Introduction

Carbohydrates, together with lipids, proteins and nucleic acids, are one of the four major classes of biologically essential organic molecules found in all living organisms.

There are a variety of interrelated classification schemes. The most useful classification scheme divides the carbohydrates into groups according to the number of individual simple sugar units. **Monosaccharides** contain a single unit; **disaccharides** contain two sugar units; and **polysaccharides** contain many sugar units as in polymers - most contain glucose as the monosaccharide unit.

Monosaccharides can be further classified by the number of carbons present. Hexoses (6-carbons) are by far the most prevalent.

Functional Groups:

Aldoses contain the aldehyde group - Monosaccharides in this group are glucose, galactose, ribose, and glyceraldehyde.

Ketoses contain the ketone group - The major sugar in this group is fructose.

These carbohydrates are classified by carbon's number and their functional groups as presented in table1.

Table 1: classification of Carbohydrates

Carbohydrates				
Monosaccharides			Disaccharides	Polysaccharides
Pentose/aldose	Hexose/ Aldose)	Hexose /ketose	Maltose	Glycogen
Ribose	Glucose Galactose Mannose Allose Altrose Idose Talsole	Fructose	Lactose	Cellulose
Lyxose		Tagatose	Sucrose (non reducing)	Starch
Xylose				
Arabinose				

Test of Carbohydrates

Molisch test:

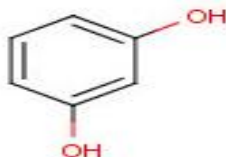
It is very important because it is a general test for the presence of carbohydrates. The molisch's reagent is α -naphtol with concentrated sulfuric acid. This test is positive for all types of carbohydrates. forming a purple or violet ring between two layers. In addition to carbohydrates, furfurals as such, some organic acids, aldehydes and ketones also give this.

Benedict's Test:

Sugars are classified as reducing or non-reducing based on their ability to act as a reducing agent during the Benedict's test. Benedict's reagent is CuSO_4 in basic medium.. The Reducing sugars include, glucose, galactose, fructose, maltose and lactose. The non-reducing sugars include sucrose and all polysaccharides are in this group. Test produces a colour change from blue to green to yellow to orange to brick red.

Seliwanoff's Test:

It is a color reaction specific for ketoses. When conce: HCl is added. ketoses undergo dehydration to yield furfural derivatives more rapidly than aldoses. These derivatives form complexes with resorcinol to yield deep red color. The test reagent causes the dehydration of ketohexoses to form 5-hydroxymethylfurfural. 5-hydroxymethylfurfural reacts with resorcinol present in the test reagent to produce a red product within two minutes. Aldohexoses reacts so more slowly to form the same product.



Resorcinol

Note: Prolonged heating will give positive test for aldohexoses which give pink color.

Barfoed's Test:

Barfoed's test is used to detect the presence of monosaccharide (reducing) sugars in solution. Barfoed's reagent, a mixture of acetic acid and copper(II) acetate, is combined with the test solution and boiled. A red copper(II) oxide precipitate is formed will indicates the presence of

reducing sugar. The reaction will be negative in the presence of disaccharide sugars because they are weaker reducing agents. This test is specific for monosaccharides .

Bial's Test:

Bial's test is used to distinguish between pentoses and hexoses. They react with Bial's reagent and are converted to furfural. Orcinol and furfural condense in the presence of ferric ion to form a colored product. Appearance of green colour or precipitate indicates the presence of pentoses and formation of muddy brown precipitate shows the presence of hexoses.

Note: Prolonged heating will give positive test for aldohexoses.

Experimental procedure

Materials:

- 1) α - naphthol / concentrated sulfuric acid.
- 2) Basic CuSO_4 solution
- 3) Hydrochloric solution.
- 4) Resorcinol
- 5) Copper acetate.
- 6) Glacial acetic acid.
- 7) Orcinol
- 8) HCl conc
- 9) FeCl_3
- 10) Different carbohydrates.
- 11)

Procedure

A) Molisch's Test

- 1- Take 1mL carbohydrate solution In a clean test tube with an angle of 45° .
- 2- Add 2-3 drops of Molisch's reagent and 0.5 mL of H_2SO_4 conc.
- 3- Down the side of the tube in a fume hood and realize the purple color.

B) Benedict's Test

- 1- Take 0.5 mL sugar solution in a clean test tube.
- 3- Add 2.5 mL Benedict's reagent.
- 2- Boil for 5 – 10 minutes and observe the formation of brick red precipitate.

C) Seliwanoff's Test

- 1- Take 0.5 mL sugar In a clean test tube.
- 2- Add 2 mL Seliwanoff's reagent.
- 3- Boil for 3 – 5 minutes then observe the bright red colored complex.

D) Barfoed's Test

- 1- Place 0.5 mL glucose solution in a cleaned test tube.
- 2- Add of 2.5 mL of Barfoed's reagent.
- 3- Boil for 10 minutes and record your observations.

E) Bail's Test

- 1- Place 0.5 mL of any pentose solution in a cleaned test tube .
- 2- Add 2.5 mL Bial's reagent.
- 3- Heat till boiling starts and observe the formation of a blue – green complex.

References:

1. Chemistry.elmhurst.edu/vchembook/541classes.html
2. www.medbiochemistry.com/molischs-test
3. Brilliantbiologystudent.weebly.com/benedicts-test-for-reducing-sugars
4. Vlab.amrita.edu/?sub=3&brch=63&sim=631&cnt=1



**Faculty of Science and Arts
Applied Chemistry
Biochemistry Lab (CHEM 266)
Experiment 9**

General tests for Carbohydrates

Report Author:

Lab Partner:

Date Due:

Purpose:

Data and Results : (Key: Color: +, No color: -)

Carbohydrate / Test					
Molisch Test					
Bendict's Test					
Seliwanoff's Test					
Barfoed's test					
Bail's Test					

Conclusion:

Errors and problems you encountered and their possible remedies: